

Lecture 23

Non-equilibrium integrability: GGE, GHD, and synthesis

Objectives. *Lecture 0* began with two experiments that refused to come to equilibrium and promised that the reason was integrability. This lecture makes the promise good. An isolated quantum system evolves unitarily, so nothing relaxes in the strict sense; what can relax is a local observable, and whether it relaxes to a thermal value depends on how much of the initial state the conserved charges remember. A generic system remembers the energy; an integrable one remembers a whole distribution of rapidities, and settles into the maximum-entropy state compatible with all the charges — the generalised Gibbs ensemble. Let that distribution vary slowly in space and it acquires a dynamics of its own, generalised hydrodynamics, at *Lecture 19*'s dressed velocity. Then the Newton's cradle, finally; then the course's dictionary, whole, with *Lecture 0*'s map reprinted and marked.

23.1 Why integrable systems do not thermalise

Prepare a system in a pure state $|\psi_0\rangle$ that is not an eigenstate of H and let it evolve. Nothing about $|\psi(t)\rangle$ relaxes: the state stays pure and, on a finite system, almost periodic. The question is about a *local* observable O (*Definition 13.2*), in the thermodynamic limit taken first. Expanding in eigenstates $H|n\rangle = E_n|n\rangle$,

$$\langle\psi(t)|O|\psi(t)\rangle = \sum_n |c_n|^2 \langle n|O|n\rangle + \sum_{m \neq n} c_m^* c_n e^{-i(E_n - E_m)t} \langle m|O|n\rangle, \quad c_n = \langle n|\psi_0\rangle. \quad (23.1)$$

Proposition 23.1 (Dephasing). *On a finite system the time average of $\langle\psi(t)|O|\psi(t)\rangle$ over $[0, T]$ converges as $T \rightarrow \infty$ to $\text{tr}(\rho_d O)$, where $\rho_d = \sum_E P_E |\psi_0\rangle \langle\psi_0| P_E$ is the initial state dephased in energy and P_E projects on the eigenspace of E . If the spectrum is non-degenerate this is the diagonal ensemble $\sum_n |c_n|^2 |n\rangle \langle n|$.*

Proof (★). The time average of $e^{-i(E_n - E_m)t}$ over $[0, T]$ is bounded in modulus by $2/(|E_n - E_m|T)$, so every term with $E_n \neq E_m$ tends to 0; the terms with $E_n = E_m$ are exactly the ones the projectors keep, and the sum is finite. ■

So whatever a local observable settles down to is fixed by ρ_d — exponentially many numbers, and no ensemble worth the name. (Not by the $|c_n|^2$ alone: inside a degenerate eigenspace the relative phases survive dephasing, and a spectator spin prepared in $|\pm x\rangle$ makes that visible.) Every relaxation statement is the claim that far fewer of those numbers matter.

The hypothesis that makes a generic system thermal

Definition 23.2 (Eigenstate thermalisation · cited: Deutsch 1991, Srednicki 1994). A Hamiltonian satisfies the *eigenstate thermalisation hypothesis* (ETH) for a local observable O if, in the bulk of the spectrum,

$$P_E O P_E = \left[O_{\text{mc}}(E/N) + \text{a fluctuation vanishing as } N \rightarrow \infty \right] P_E, \quad (23.2)$$

with O_{mc} a smooth function — the microcanonical average at that energy density. On a non-degenerate spectrum this is the familiar $\langle n | O | n \rangle = O_{\text{mc}}(E_n/N) + \dots$; where energies are degenerate it has to be asked of the whole block, or a spectator degree of freedom defeats it with every diagonal element right.

If (23.2) holds and $|\psi_0\rangle$ has subextensive energy fluctuations, Proposition 23.1 averages a smooth function over a narrow window and the answer is the microcanonical one: the system *thermalises*, in the only sense a closed quantum system allows. Nothing has been proved — ETH is a hypothesis about a Hamiltonian, checked numerically model by model, and (23.2) is exactly the statement a figure can test (Figure 23.1).

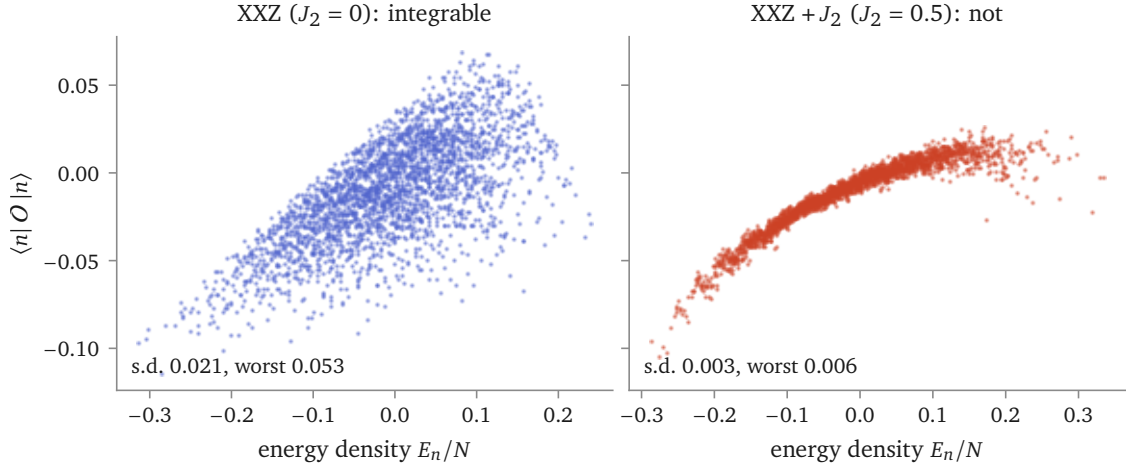


Figure 23.1: Eigenstate expectation values of the two-site operator $O = S_1^z S_2^z$ — equal, on the momentum eigenstates drawn, to the density $\frac{1}{N} \sum_n S_n^z S_{n+1}^z$, which is what is computed — in every state of one symmetry sector ($N = 18$, $S^z = 1$, $k = \pi/3$) of (13.10) at $\Delta = 0.55$: Lecture 13’s two chains, seen now through an observable. Left, integrable, a cloud; right, with $J_2 = 0.5$, a curve. *The two numbers say different things.* The *standard deviation* in a narrow energy window measures a typical eigenstate, and it falls in *both* models — for XXZ like $N^{-1/2}$ ($\text{sd}\sqrt{N} = 0.091, 0.092, 0.091$ at $N = 14, 16, 18$). *Weak* eigenstate thermalisation holds. The *maximum* deviation measures the worst eigenstate, and there they part: $0.075 \rightarrow 0.078 \rightarrow 0.083$ for XXZ, *growing*, against $0.028 \rightarrow 0.031 \rightarrow 0.012$ for $J_2 = 0.5$. Rare eigenstates stay far from the local average — and a quench does not sample eigenstates typically. Symmetry resolution matters for the reason it did in § 13.4.

The charges are the memory

The left panel is not an accident of the observable. An integrable chain has $\sim N$ local charges $Q_1 = H, Q_2, Q_3, \dots$ (§ 13.3, Theorem 13.4), all commuting, so its eigenstates can be labelled by all of them at once. What is expected to replace (23.2) there is a *generalised* eigenstate thermalisation: $\langle n | O | n \rangle$ a smooth function of *all* the charge densities · cited: *Caux and Essler 2013*. That reduces to a function of the energy alone only if the others are fixed by it, and they are not — two eigenstates at the same energy density generically carry different Q_3 . The charges remember which state the system was prepared in, and a local observable can see the memory.

Reading

Remark 23.3 (What is *not* being claimed). Three things. An initial state that happens to carry a thermal state's charge densities relaxes to Gibbs values, because the ensemble of § 23.2 is then thermal. Some observables cannot tell the two apart even when the state can (Remark 23.9). And ETH is a theorem in neither direction: every statement here about a non-integrable chain is numerical.

23.2 The generalised Gibbs ensemble, and a quench done in full

Definition 23.4 (Generalised Gibbs ensemble · cited: *Rigol, Dunjko, Yurovsky and Olshanii 2007*). Given conserved charges Q_a and their initial expectation values $\langle \psi_0 | Q_a | \psi_0 \rangle$, the *generalised Gibbs ensemble* is the maximum-entropy state reproducing all of them,

$$\rho_{\text{GGE}} = \frac{1}{Z} \exp \left(- \sum_a \beta_a Q_a \right), \quad (23.3)$$

the multipliers β_a being fixed by those expectation values.

With one charge, $Q_1 = H$, this is the Gibbs state and $\beta_1 = 1/T$; the question is what the others do. The claim to be tested — that a local observable relaxes to its ρ_{GGE} average — can be settled by calculation on Lecture 14's chain.

The quench

Take the transverse-field Ising chain (14.1) at coupling J , start in the ground state at field h_0 , and at $t = 0$ change the field to h . Both Hamiltonians are quadratic in the *same* fermions c_k : only the Bogoliubov angle moves. Write θ_k for the post-quench angle of (14.16), θ_k^0 for the pre-quench one, and

$$\chi_k = \theta_k - \theta_k^0, \quad \cos 2\chi_k = \frac{a_k a_k^0 + b_k^2}{\varepsilon_k \varepsilon_k^0}, \quad \sin 2\chi_k = \frac{4J(h_0 - h) \sin k}{\varepsilon_k \varepsilon_k^0}, \quad (23.4)$$

with $a_k = 2(h - J \cos k)$, $b_k = 2J \sin k$, $\varepsilon_k = \sqrt{a_k^2 + b_k^2}$ as in § 14.4, and a_k^0, ε_k^0 the same at h_0 . The rotation taking pre-quench modes to post-quench ones is the rotation by the *difference* angle, $\gamma_k = \cos \chi_k \gamma_k^0 - i \sin \chi_k (\gamma_{-k}^0)^\dagger$, and the pre-quench ground state is the γ^0 vacuum.

Proposition 23.5 (The occupations are the conserved data). *After the quench the post-quench mode occupations are*

$$n_k = \langle \psi(t) | \gamma_k^\dagger \gamma_k | \psi(t) \rangle = \sin^2 \chi_k, \quad (23.5)$$

independent of t , and even in k .

Proof. Each $\gamma_k^\dagger \gamma_k$ commutes with $H(h)$, so its expectation is constant; its value in the γ^0 vacuum is $|\sin \chi_k|^2$ by the rotation above. Evenness: $\cos 2\chi_k$ in (23.4) depends on k only through $\cos k$. ■

The difference angle is not a bookkeeping convention: the sum angle gives a well-formed, plausible and wrong n_k , off by as much as 0.48.

What a local observable does

The transverse magnetisation $m^z = \frac{1}{N} \sum_n \sigma_n^z$ is the density of a one-site operator, and in a translation-invariant state it is $\langle \sigma_1^z \rangle$ exactly; (14.5) makes it $1 - \frac{2}{N} \sum_k c_k^\dagger c_k$.

Theorem 23.6 (Relaxation to the generalised Gibbs ensemble). *Let $J, h, h_0 > 0$ with $h \neq J$ and $h_0 \neq J$. For the quench above on a ring of N sites, with the NS momenta of the even sector,*

$$m^z(t) = \frac{1}{N} \sum_k \left[\cos 2\theta_k \cos 2\chi_k + \sin 2\theta_k \sin 2\chi_k \cos(2\varepsilon_k t) \right]. \quad (23.6)$$

In the thermodynamic limit the sum becomes $\int dk/2\pi$, the oscillating term tends to 0 as $t \rightarrow \infty$ like $t^{-3/2}$, and the limit is

$$m^z(\infty) = \int \frac{dk}{2\pi} \cos 2\theta_k (1 - 2n_k) = \text{tr}(\rho_{\text{GGE}} m^z), \quad (23.7)$$

the average in the ensemble (23.3) built from the occupations (23.5).

Proof. The ground state at $h_0 > 0$ is in the even sector — in the σ^x basis the field gives non-positive off-diagonal elements connecting every configuration, so Perron–Frobenius makes the ground vector unique and strictly positive, and Π only permutes those configurations — so the NS grid is the right one. Each pair $(k, -k)$ then spans a two-dimensional even-parity space, and the expectation of $c_k^\dagger c_k$ in it is $\frac{1}{2}(1 - \cos 2\theta_k \cos 2\chi_k) - \frac{1}{2} \sin 2\theta_k \sin 2\chi_k \cos 2\varepsilon_k t$; summing gives (23.6). Every term is even in k , since $\cos 2\theta$ and $\cos 2\chi$ are even and $\sin 2\theta$ and $\sin 2\chi$ are odd.

For the limit, $\varepsilon'(k) \propto \sin k$ vanishes exactly at $k = 0$ and $k = \pi$ and nowhere else, and the amplitude is

$$\sin 2\theta_k \sin 2\chi_k = \frac{8J^2(h_0 - h) \sin^2 k}{\varepsilon_k^2 \varepsilon_k^0}. \quad (23.8)$$

This is where the hypotheses are spent. With $h \neq J$ and $h_0 \neq J$ neither ε vanishes anywhere, so (23.8) has a double zero at both stationary points, and stationary phase with an amplitude vanishing to second order gives $t^{-3/2}$ — not the generic $t^{-1/2}$ of a non-vanishing amplitude. What is left is $\int \cos 2\theta_k \cos 2\chi_k$, and $\cos 2\chi_k = 1 - 2n_k$ by (23.5). The ensemble (23.3) built from the charges $Q[f]$ below is diagonal in the occupations, so its modes are independent with occupations n_k and the anomalous correlator $\langle \gamma_k^\dagger \gamma_{-k}^\dagger \rangle$ vanishes; the two numbers agree. ■

Remark 23.7 (Both hypotheses are needed, and so is $h > 0$). Three boundaries of Lecture 14's domain $J, h \geq 0$ are excluded above, each failing differently. At $h_0 = J$ the factor $\varepsilon_k^0 \sim 2J|k|$ cancels one power of $\sin^2 k$ in (23.8): the zero is simple and the decay is t^{-1} , of coefficient $1/8\pi$ at $J = h_0 = 1$, $h = 2$ (measured -1.002). At $h = 0$ the dispersion is *flat*, there is no stationary phase, and the oscillation never decays — envelope 0.42 at $t \approx 500$. At $h_0 = 0$ the integral is fine but there is no longer a ground state: the cats $(|+x\rangle^{\otimes N} \pm |-x\rangle^{\otimes N})/\sqrt{2}$ are degenerate, and (23.6) is the even one — at $N = 4$, $h = 2$, $t = \frac{1}{2}$, 0.4707 against the odd cat's 0.2476. The fourth excluded value, $h = J$, is no failure of the decay at all: the exponent survives there (measured -1.495), from $k = \pi$ alone, $k = 0$ being then a corner of ε and not a stationary point. What $h = J$ costs is the *comparison*: there $\cos 2\theta_k = \varepsilon_k/4J$, so m^z is proportional to the energy and the two ensembles agree whatever the state — an ensemble comparison is only as good as its observable (the reading).

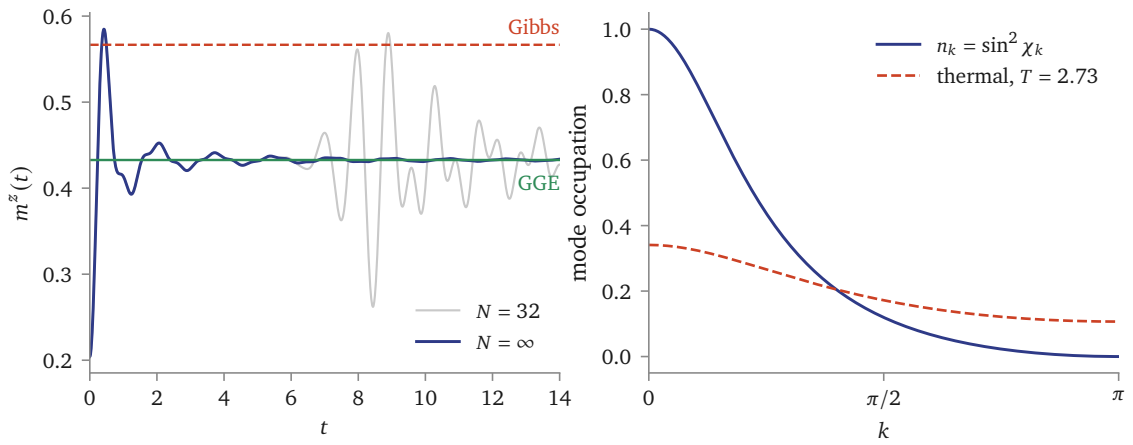


Figure 23.2: The quench $J = 1$, $h_0 = 0.4 \rightarrow h = 1.9$. *Left*: $m^z(t)$ from (23.6). In the thermodynamic limit (dark) it settles onto the generalised Gibbs value 0.4327; the Gibbs value at the temperature $T = 2.730$ carrying the *same energy* is 0.5667. At $N = 32$ (pale) nothing decays — the sum revives, and only its time average converges. *Right*: the conserved $n_k = \sin^2 \chi_k$ against the thermal occupations of the same energy; the closest Fermi–Dirac fit, at $T = 3.24$, is out by 0.243 root mean square. The left panel's formula is checked against exact diagonalisation of the spin chain at $N = 8, 10$, the initial state's parity included.

Remark 23.8 (The order of the limits). Theorem 23.6 takes $N \rightarrow \infty$ first and $t \rightarrow \infty$ second, and the two do not commute. At finite N (23.6) is a finite sum of cosines: almost periodic, and it revives, as the pale curve of Figure 23.2 does at $t \approx 8$. What converges there is the *time average* — each $\cos 2\varepsilon_k t$ averages to $O(1/\varepsilon_k T)$, so the average tends to the same number like $1/T$. (Proposition 23.1 gives the limit; the cosines give the rate.) Every experiment is a finite system watched for a finite time.

Remark 23.9 (An observable that cannot tell the ensembles apart). Quench to the critical field $h = J$ and the two predictions for m^z agree exactly, for every h_0 — and not because anything thermalised. There $a_k = 4J \sin^2(k/2)$ and $\varepsilon_k = 4J |\sin(k/2)|$, so $\cos 2\theta_k = \varepsilon_k/4J$ and (23.7) makes m^z proportional to the energy, which the Gibbs ensemble is built to reproduce. An ensemble comparison is only as good as its observable.

Which combinations of the n_k are local

Definition 23.4 asks for charges, and a single n_k is not an observable of the chain: it is a sum over all N sites whose coefficients never decay. § 14.6 built the local charges by weighting the k -block of (14.14); this is that weighting.

Proposition 23.10 (The local charges of the Ising chain · sketched). Let $J, h > 0$ with $h \neq J$, and let f be an even function on the Brillouin zone. Write $Q[f] = \sum_k f(k)(n_k - \frac{1}{2})$. Then $Q[f]$ is a sum of fermion bilinears whose real-space coefficients are the Fourier modes of $f \cos 2\theta$ and $f \sin 2\theta$. Hence

$$Q[f] \text{ has finite range } \iff f = \varepsilon \cdot (\text{a trigonometric polynomial}), \quad (23.9)$$

and $f = \varepsilon \cos mk$ gives a charge of range $m + 2$ in the spins.

Proof sketch (★). In the pair basis $Q[f] = \sum_{0 < k < \pi} \Psi_k^\dagger (f h_k / \varepsilon_k) \Psi_k$, so the hopping and pairing functions are $A = f a_k / \varepsilon_k$ and $B = f b_k / \varepsilon_k$. Finite range means both are trigonometric polynomials, and a_k, b_k are. For the converse, $g = f / \varepsilon$ is recovered from A and B by one identity,

$$(h + J \cos k) a_k - J \sin k b_k = 2(h^2 - J^2), \quad \text{so} \quad g = \frac{(h + J \cos k)A - J \sin k B}{2(h^2 - J^2)}, \quad (23.10)$$

a trigonometric polynomial whenever A and B are — and this is exactly where $h \neq J$ is spent. Taking $f = \varepsilon$ returns H itself; $f = \varepsilon \cos k$ returns $-\frac{1}{2}Q_+$ of (14.23) up to a constant, checked against the even-sector ground state at $N = 8, 10$, where the sign is visible. ■

The ε weight is not decoration. Drop it, and $Q[\cos mk]$ is not local: its coefficients decay exponentially, at the rate $|\ln(h/J)|$ — the branch points of $1/\varepsilon$ sit at $z = e^{ik} = h/J$ and $z = J/h$, and whichever is nearer the unit circle sets the rate — without ever vanishing. Such a charge is *quasi-local*, the class § 13.3 flagged and did not need and § 23.3 does. Both ends of the range are special: at $h = J$ the two branch points meet on the circle and the coefficients decay only like $1/d$, while at $h = 0$, where $\varepsilon = 2J$ is constant, the weight is a number and every trigonometric polynomial is strictly local again.

None of this weakens Theorem 23.6: knowing every $Q[\varepsilon P]$ is knowing every Fourier coefficient of $\varepsilon(k)n(k)$, and $\varepsilon > 0$ almost everywhere, so the strictly local charges fix n_k almost everywhere and with it every local observable's relaxed value. Numerically, keeping $Q[\varepsilon \cos mk]$ for $m \leq M$ and maximising the entropy gives m^z to 1.7×10^{-3} at $M = 4$ and 8×10^{-8} at $M = 16$ — and at $M = 0$, where only the energy is kept, that is the Gibbs ensemble, with exactly the error of Figure 23.2.

Remark 23.11 (This was known in 1970). Barouch, McCoy and Dresden computed the magnetisation of the XY chain after a field quench in 1970 and found it non-ergodic: it does not approach the equilibrium value. The ensemble that explains it was written down thirty-seven years later.

23.3 Interacting models: one function carries the state

Free fermions made the argument exact, and easy, because the charges were quadratic. The replacement for an interacting model is already built: a macrostate in the thermodynamic limit is a root density $\rho(\lambda)$, or the filling fraction $\vartheta = \rho/\rho_t$ of Definition 19.8, and every extensive charge is

$$\frac{\langle Q_a \rangle}{L} = \int q_a(\lambda) \rho(\lambda) d\lambda, \quad (23.11)$$

with q_a the charge's one-particle eigenvalue — $q = k^2$ for the energy of the gas, $q = k$ for its momentum. A family of charges fixes ρ exactly when its q_a separate root densities; given such a *complete* family, the generalised Gibbs ensemble of Definition 23.4 is a *single function* — for a one-species gas; one filling per quasiparticle species in general, as Lecture 20's strings and Lecture 22's two families already needed — the same ϑ that Lecture 20 minimised the free energy over, now fixed by the initial state instead of by a temperature. Maximising (20.1) at fixed (23.11) is the generalised thermodynamic Bethe ansatz, and it returns (20.4) in *dimensionless* form: with $\eta = \ln(\rho_h/\rho)$ and $w = \sum_a \beta_a q_a$,

$$\eta(\lambda) = w(\lambda) - \frac{1}{2\pi} \int K(\lambda - \mu) \ln(1 + e^{-\eta(\mu)}) d\mu, \quad \vartheta = \frac{1}{1 + e^\eta} \quad (23.12)$$

· *cited*. The multipliers carry the temperature, so (20.4)'s explicit T 's go: at thermal equilibrium $\eta = \varepsilon/T$, and $w = (\varepsilon_0 - \mu)/T$ returns Lecture 20's equation exactly. *Replacing only the source in (20.4) and keeping its T would halve the exponent* — it gives $\vartheta = (1 + e^{k^2/4})^{-1}$ where the Gibbs state at $T = 2$ has $(1 + e^{k^2/2})^{-1}$.

Two things then have to be said honestly.

There are two routes to ϑ , with different inputs. The one above needs a complete family of charges and their initial values, and nothing else. The *quench action* takes the other: it writes the time evolution as a saddle point over macrostates using the overlaps $\langle n|\psi_0 \rangle$, and gets both the stationary state and the approach to it · *cited: Caux and Essler 2013*. Overlaps are not an extra ingredient the first route needs; they are what the second uses instead of completeness.

The ultralocal charges are not enough. For the XXZ chain the charges generated by $t(\lambda)$ as in § 17.5 do *not* fix ϑ : two macrostates with different ϑ can share all of them, and the quasi-local charges of § 13.3 — the ones whose densities decay exponentially instead of stopping, exactly as $Q[\cos mk]$ did above — are needed to close the ensemble · *cited: Ilievski, De Nardis, Wouters, Caux, Essler and Prosen 2015*. This is where the free-fermion picture misleads if pushed: there the strictly local charges sufficed, because ε never vanished.

Remark 23.12 (More than one family). Lecture 22's gases carry two coupled families of rapidities, charge and spin. Nothing above changes in form: the macrostate is a pair $(\rho, \rho_{\text{spin}})$, the charges are integrals against both, and the generalised ensemble is a pair of filling fractions coupled through the nested equations (22.16)–(22.17). What changes is that the spin family carries no energy of its own (Theorem 22.7: $E = \sum k_j^2$), so the energy does not *determine* how the spin rapidities are distributed. It does not follow that the energy says nothing about them: the spin roots enter (22.16) and move the charge roots, so fixing the energy can exclude whole spin sectors — at $N = 2, L = c = 1$ the zero-momentum singlet of Example 22.8 has $E = 1.8439$ while every triplet state has $E \geq 4\pi^2$ — and a thermal saddle couples the two families. What a generalised ensemble can do is be *told* a spin distribution that no temperature would have chosen.

23.4 Generalised hydrodynamics

Now let the state vary in space: $\vartheta(\lambda, x, t)$, slowly on the scale of the interactions — a *local* generalised Gibbs ensemble. That is the assumption, and the only one; there is no gradient expansion beyond leading order, which is what *Euler scale* means.

How fast does a quasiparticle actually move?

Lecture 9 found that a collision permutes the momenta and leaves a *displacement* behind (Theorem 9.9, § 9.7), and Lecture 16 found the same shift quantum mechanically. A quasiparticle crossing a gas therefore collects one displacement per particle it passes, and does not travel at its bare velocity.

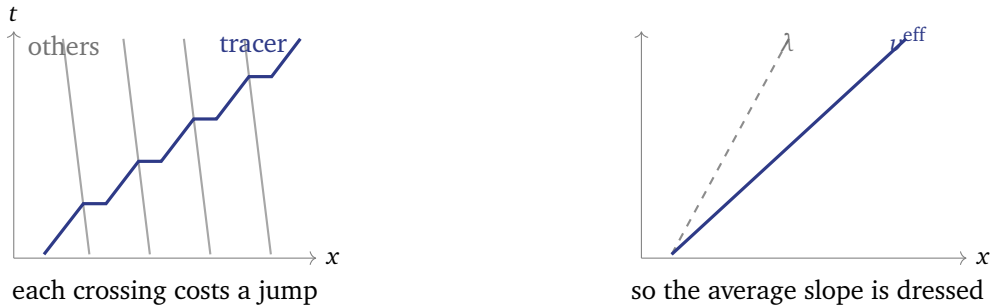


Figure 23.3: The effective velocity. *Left:* a tracer (dark) crossing a gas; each crossing displaces it, here forwards. *Right:* over many crossings the bare slope λ becomes an average slope v^{eff} , which depends on the whole state through the rate at which crossings happen. For hard rods the bookkeeping closes in one line, (23.13); in general into Lecture 19's dressing equation, (23.14).

Proposition 23.13 (Hard rods · sketched). Let the gas be N identical rods of length a and unit mass on a line, moving freely between elastic collisions, so that the bare momentum is $p_0(\lambda) = \lambda$ and the bare energy $\varepsilon_0(\lambda) = \lambda^2/2$. Equal masses exchange velocities, so the set of velocities is only permuted and the gas is integrable; its rapidities are the velocities. A

tracer of velocity λ jumps forward by a each time it passes a rod, so

$$v^{\text{eff}}(\lambda) = \lambda + a \int d\alpha \rho(\alpha) [v^{\text{eff}}(\lambda) - v^{\text{eff}}(\alpha)] = \frac{\lambda - aJ}{1 - an}, \quad n = \int \rho, \quad J = \int \alpha \rho. \quad (23.13)$$

Proof. The rate at which the tracer passes rods of velocity α is $\rho(\alpha) |v^{\text{eff}}(\lambda) - v^{\text{eff}}(\alpha)|$ and the jump is $\pm a$ with the sign of the relative velocity, which is the first equality. Integrating it against ρ gives $\int \rho v^{\text{eff}} = J$ identically, and substituting that back solves it. ■

Equation (23.13) is Equation (19.13) with the *constant* kernel $K = -a$ — not a local one, unlike Lecture 21's delta, which would dress every function by the same factor and cancel in the ratio. Dressing with it gives $2\pi\rho_t = (1)^{\text{dr}} = 1 - an$, the free volume fraction, and $(\varepsilon'_0)^{\text{dr}} = \lambda - aJ$, whose ratio is (23.13). Lecture 20's $1/c$ row says this is no curiosity: at large c the Lieb–Liniger gas is free fermions in a box *lengthened* by $2/c$ per particle, that is, a hard-rod gas of length $a = -2/c$. In general

$$v^{\text{eff}}(\lambda, x, t) = \frac{(\varepsilon'_0)^{\text{dr}}}{(k'_0)^{\text{dr}}}, \quad (23.14)$$

with the dressing of (19.13) at the local ϑ and k_0 the excitation's *own* bare momentum (Remark 19.10). Write it this way and not as $\varepsilon'/2\pi\rho_t$, which is Equation (20.7). Lecture 20 got that by differentiating a *thermal* Yang–Yang equation driven by $\varepsilon_0 - \mu$; differentiating (23.12) gives $\eta' = (w')^{\text{dr}}$ instead, which dresses the *combination* w and not the physical energy. (Free fermions with $w = \lambda^4$: there $2\pi\rho_t = 1$ and $\eta' = 4\lambda^3$, against 2λ from $\varepsilon_0 = \lambda^2$.) The two coincide at thermal equilibrium, where $\eta = \varepsilon/T$ and the T cancels from the ratio — which is where Lecture 20 used it, saying explicitly that calling the ratio a propagation speed was not proved there. This section is where it is used, and the rods are the reason to believe it.

The equation

$$\partial_t \rho(\lambda, x, t) + \partial_x [v^{\text{eff}}(\lambda, x, t) \rho(\lambda, x, t)] = 0 \quad \text{for every } \lambda. \quad (23.15)$$

One continuity equation per rapidity · *sketched*: Castro-Alvaredo, Doyon and Yoshimura 2016; Bertini, Collura, De Nardis and Fagotti 2016: the number of quasiparticles of each rapidity is separately conserved — what integrability buys — and they are transported at the dressed velocity, which is what interaction costs. Ordinary hydrodynamics has five such equations in three dimensions; this has one per point of a continuum.

Remark 23.14 (Which way the flow runs). Let g_t be the flow of $\dot{x} = v(x)$. The solution of $\partial_t \rho + \partial_x (v\rho) = 0$ is the *pushforward*

$$\rho_t(x) = \rho_0(g_{-t}x) |\det Dg_{-t}(x)|, \quad (23.16)$$

with g_{-t} and with the Jacobian. Both are easy to lose and neither is cosmetic: $\rho_0(g_t x) |\det Dg_t|$ solves the same equation with $t \rightarrow -t$, and dropping the Jacobian solves neither. When v does not depend on x — the free case — (23.16) is just $\rho_0(x - vt)$, with the minus. In (23.15) the velocity depends on the solution and so on the time, and then g_t is not a one-parameter group: what the formula needs is the *inverse* of the map carrying time 0 to time t , which is no longer g_{-t} . For $v(x, t) = t$ both g_t and g_{-t} are $x \mapsto x + t^2/2$ while the inverse is $x \mapsto x - t^2/2$, and only the inverse solves the equation. What (23.16) fixes is the *direction*, and that does not change.

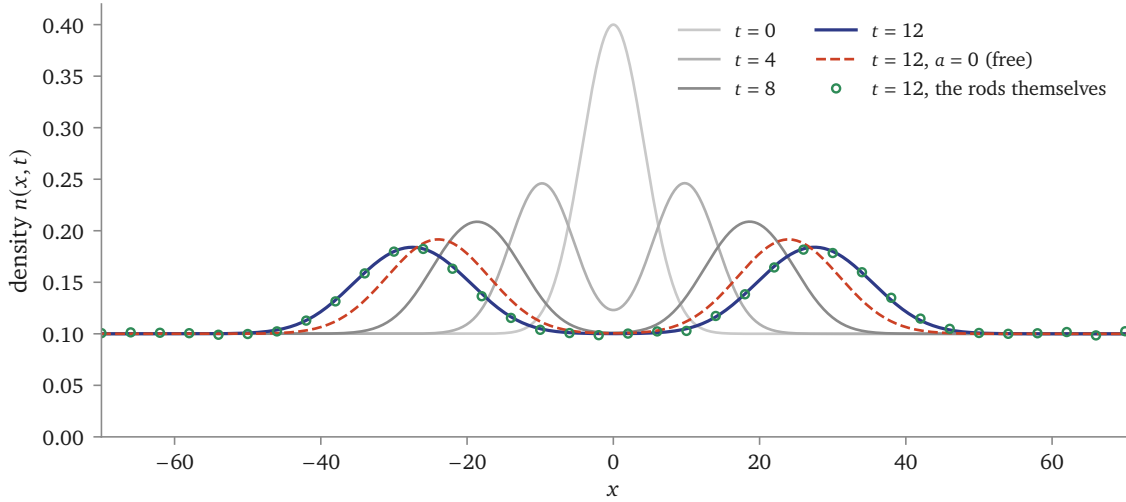


Figure 23.4: A lump of hard rods splitting, from (23.15): an excess of rods on a uniform background (palest) with rapidities near $\pm\lambda_0$ separates into a left-mover and a right-mover — the Newton’s cradle with one collision instead of thousands. The rods have $a = 1$ and an reaches 0.40, so the dressing is large: the dashed curve is the *same* initial state evolved with $a = 0$, its peak 3.5 units behind. The circles are a direct simulation of the rods — no hydrodynamics — prepared as a genuine non-overlapping configuration and lying on the GHD curve to 3% of the lump height. The check script verifies that the solver conserves the number of rods to 10^{-12} , separately in every rapidity slice.

Figure 23.4 is the theory in one picture, and the one place in this course where an Euler-scale claim is tested against the microscopic model it describes: the circles are rods, evolved exactly, not a second solution of (23.15). The agreement is a law of large numbers and needs many particles per hydrodynamic cell — at one rod per ten units of length the same comparison fails outright.

23.5 The Newton’s cradle, finally

Equation (23.15) has an equivalent form: the filling fraction is not conserved but *advected*, $\partial_t \vartheta + v^{\text{eff}} \partial_x \vartheta = 0$ in the homogeneous case — not obvious, since v^{eff} depends on ϑ · *cited*: Castro-Alvaredo, Doyon and Yoshimura 2016. There a trapping potential $V(x)$ enters as a force on the rapidity, and the content is that quasiparticles follow characteristics:

$$\partial_t \vartheta + v^{\text{eff}} \partial_x \vartheta - \partial_x V \partial_\lambda \vartheta = 0, \quad \text{i.e.} \quad \dot{x} = v^{\text{eff}}, \quad \dot{\lambda} = -\partial_x V, \quad (23.17)$$

with ϑ constant along them · *cited*: Doyon and Yoshimura 2017. In Lecture 16’s units, $\hbar = 2m = 1$, the rapidity is the momentum: λ obeys Newton’s second law and x a dressed version of the first.

One caution first: (23.17) does not say the trapped model is integrable. A position-dependent potential breaks the homogeneous charges, no Bethe ansatz solves the trapped Lieb–Liniger Hamiltonian, and (23.17) is an Euler-scale approximation, good while V varies slowly on the scale of the interactions · *cited*: Doyon and Yoshimura 2017; Caux et al. 2019, App. H. It says two things.

First, at the two ends of the coupling range the trap costs nothing at all. There $v^{\text{eff}} = 2\lambda$ — the ideal gas, and the impenetrable limit, where the gas carries the free fermions’ energy

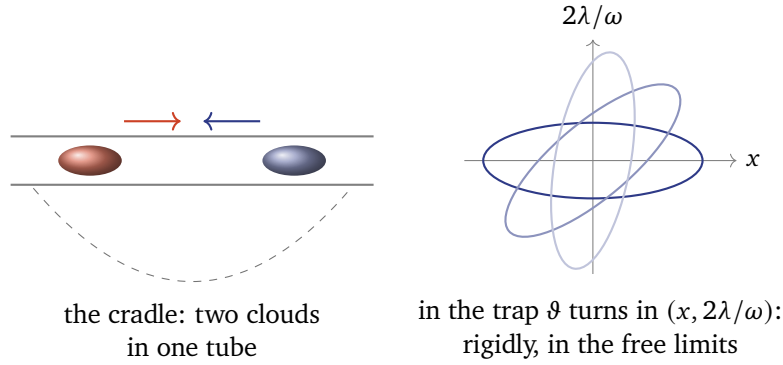


Figure 23.5: [Lecture 0](#)’s cradle ([Figure 2](#)), and what generalised hydrodynamics says about it. A harmonic trap adds the force term of (23.17). *In the two free limits* the characteristics are linear: the distribution turns rigidly in the scaled plane $(x, 2\lambda/\omega)$ at the trap frequency and returns to itself each period, so the clouds pass through each other and re-form, thousands of times. At intermediate coupling v^{eff} depends on the local state, the turning is not rigid and the return is only approximate. The free skeleton of the mechanism, not a claim about every coupling.

exactly ([Proposition 19.7](#)) — and (23.17) is a *linear* flow of period $2\pi/\omega$: the distribution turns rigidly on ellipses of fixed $x^2 + (2\lambda/\omega)^2$ and returns exactly to itself ([Figure 23.5](#)). That is the cradle. In between, v^{eff} depends on the local state, the return is only approximate and the shape drifts — which is what the experiments see. Second, $\int \rho \, dx$ is *not* conserved in a trap, since (23.17) moves λ : “the momentum distribution does not relax” means a shape returning, not a constant.

The theory, and then the data. Caux, Doyon, Dubail, Konik and Yoshimura applied GHD to the cradle geometry in 2019, at a coupling close to the original experiment’s. The first experimental test is Schemmer, Bouchoule, Doyon and Dubail’s atom chip: one cloud after a longitudinal trap quench, where GHD works and conventional hydrodynamics, which assumes local *thermal* equilibrium, does not. The quantitative test is Malvania, Zhang, Le, Dubail, Rigol and Weiss, in the same laboratory’s gases, measuring rapidity distributions directly through dozens of trap oscillations at $\gamma = c/n$ from 0.3 to 9.3 · *cited*. **There is, as of writing, no published comparison of generalised hydrodynamics with the 2006 data itself**; theory and experiment meet in the later apparatus, not in the original plot.

What else spoils it. The atoms of [Lecture 0](#)’s tube are not exactly one-dimensional: virtually exciting a transverse mode and coming back leaves an effective *three*-body collision along the tube, which no Bethe ansatz solves, and whose rate sets a thermalisation time · *cited*: *Mazets, Schumm and Schmiedmayer 2008*. Integrability is therefore *approximate* on two counts, and the question is one of time scales.

Remark 23.15 (Prethermalisation, and the quantum KAM question). Under weak integrability breaking a system started far from equilibrium first relaxes, on the fast scale, to the generalised ensemble of the *unbroken* model — a *prethermal* plateau — and only much later drifts to the thermal state, on a scale that grows as the breaking is reduced · *cited*: *Bertini, Essler, Groha and Robinson 2015*; *observed in Gring et al. 2012*. This is the nearest quantum relative of [Theorem 6.8](#), and the analogy is honest about where it stops: KAM is a theorem, with hypotheses, about which tori survive; prethermalisation is a statement about time

scales with no theorem behind it. Whether approximate charges survive weak breaking in a controlled sense is the *quantum KAM* question, and it is open — cited: Brandino, Caux and Konik 2015.

23.6 Synthesis

The dictionary, whole

Every Part II lecture carried a Dictionary box; here they are together, typed. **T** is a theorem of this course — an explicit proved map or $\hbar \rightarrow 0$ limit between the columns; **A** a structural analogy, one role played by two separately established facts; **C** a citation.

Part I	Part II	L	type
Poisson bracket $\{\cdot, \cdot\}$	commutator $[\cdot, \cdot] / i\hbar$	13	A
n integrals in involution	no agreed definition; criteria	13	A
Poincaré sections	level statistics	13	A
linear action–angle system	free fermions, $\sum \varepsilon(k)n_k$	14	A
$\oint p \cdot dq = 2\pi\hbar n$	Bethe equations, logarithmic form	15	A
permuted momenta, braid	factorised S-matrix, Yang–Baxter	16	A
classical displacement	wave-packet displacement $\mp\theta'$	16	A
Lax matrix $L(\lambda)$	Lax operator $L_n(\lambda)$	17	T
Sklyanin bracket	RTT	17	T
$r(\lambda)$	$R/\lambda = 1 + i\hbar r + O(\hbar^2)$	17	T
$\{t(\lambda), t(\mu)\} = 0$	$[t(\lambda), t(\mu)] = 0$	17	T
$\text{tr } L^k$	$d^k \ln t(\lambda) _0$	17	A
classical Yang–Baxter	Yang–Baxter	17	T
spectral curve $\det(z - T)$	T–Q relation	18	A, C
bands and gaps	Fermi sea of rapidities	19	A
frequency $\partial K / \partial I$	dressed velocity	19	A
measure $dI d\varphi$	Yang–Yang entropy	20	A
CM by reduction; isochrony	CS; equally spaced trapped spectrum	21	A
$\mathfrak{sl}_N$ roots	Dynkin-indexed nested equations	22	A, C
elastic non-diffractive scattering	labels transported by X	22	A
invariant tori	generalised Gibbs ensemble	23	A
KAM	prethermalisation	23	A, C

Five rows are theorems, and all five are [Lecture 17](#)’s. Four are the quantum Toda chain; the fifth — the classical Yang–Baxter equation as the order- $\hbar^2$ part of the quantum one — is proved in [Lecture 17](#)’s reading tier. Say the count aloud rather than let a reader find it. It is no weakness of the subject: $R = 1 + i\hbar r + O(\hbar^2)$ is a genuine deformation, and the table records what it does to objects whose classical and quantum versions are separately constructed. The thesis of this course is that the table is the subject — not that every row is a limit.

Where the course sat, and what is next to it

Dictionary

The invariant tori of [Theorem 4.8](#): the motion is confined to the level set of the integrals, and carries the initial values of all of them for ever (export E2) $\longrightarrow$ the generalised Gibbs ensemble ([23.3](#)): a local observable relaxes, but only to the maximum-entropy

	finutely many degrees of freedom	infinitely many (fields)
quantum	Quantum mechanics. Criteria, not a definition (L13); free fermions (14); coordinate and algebraic Bethe ansatz (15–18); thermodynamics (19–20); Dunkl operators (21); nesting (22); out of equilibrium (23). Part II, done.	Quantum field theory in two dimensions. Factorised S -matrix, form factors, thermodynamic Bethe ansatz — the same Equation (20.4), with relativistic dispersion. Not covered; the nearest frontier.
classical	Classical mechanics. Liouville–Arnold (L1–5); when it fails (6); Lax pairs and r -matrices (7–8); Toda (9–10); Calogero–Moser and reduction (11–12). Part I, done.	Classical field theory. Zero curvature, solitons, the inverse scattering transform; § 10.5’s spectral curve is the finite-dimensional end of the data that builds this column’s finite-gap solutions. Not covered.

Figure 23.6: Figure 7, marked. The course occupied the left column and did both of its rows: twelve lectures below, eleven above, with the dictionary running between them. The right column was excluded by the scope contract, not by difficulty — and both of its cells are reached from the left by the same constructions.

state carrying the initial values of all the charges — one number per charge instead of one torus (*type A, structural analogy; the Part II side proved here for the Ising quench, Theorem 23.6*)

KAM (Theorem 6.8): under a small perturbation most tori survive, and the motion *on* them stays quasi-periodic for ever — with resonant gaps in between, which the theorem does not control → prethermalisation: under weak integrability breaking the generalised ensemble survives as a long-lived plateau, and the system thermalises only on a much longer scale (*type A, cited; the quantum statement is about time scales and there is no theorem behind it — Remark 23.15*)

Reading

23.7 The frontier, and reduction as a construction principle

Four directions lead out of this course. **Classification.** Which $R(\lambda)$ solve the Yang–Baxter equation? Classically there is a theorem: Belavin and Drinfeld sorted the non-degenerate solutions of (8.13) for a simple Lie algebra into rational, trigonometric and elliptic families. Quantum mechanically no list is closed, even for a two-dimensional auxiliary space and even of difference form, and Lecture 22 met an R of no difference form at all. The systematic language is quantum groups: Yangians and affine quantum algebras, whose representations are the R -matrices. **Separation of variables.** Sklyanin’s method replaces the Bethe ansatz by a change of variables to the eigenvalues of $B(\lambda)$ — and § 18.7 found B nilpotent on the homogeneous chain, so there are none to change to and the construction must twist the monodromy first. (The pseudovacuum was never the problem: Lecture 18 built it.) It is where Lecture 10’s spectral curve $\det(z - T(\lambda))$ returns, as a quantum spectral curve. **Beyond Euler.** Equation (23.15) is the leading order; diffusive corrections, and the anomalous transport of the isotropic Heisenberg chain, are current work. **Integrable circuits.** Discrete-

time quantum circuits built from solutions of Yang–Baxter are integrable models in which everything here can be recomputed, and are the form in which the subject has reached quantum simulators.

And one principle, which [Lecture 12](#) promised would return. It built the Calogero–Moser system out of free motion on $\text{Herm}(N)$, a symmetry, a moment-map constraint and a gauge fixing. Reduction returns in three places, all of them beyond this course.

Quantum Hamiltonian reduction. The classical recipe has an operator version: free motion becomes the Laplacian on a group, the constraint becomes a representation-theoretic condition, and the radial part of the Laplacian is the Calogero–Sutherland operator of [Lecture 21](#). This is why the Dunkl operators of § 21.3 look like covariant derivatives: they are.

Nested systems from root systems. [Lecture 22](#)’s $A_{\kappa-1}$ label is the visible end of the same structure: the Calogero–Moser models are indexed by root systems ([Lecture 12](#), reading), and so are the nested Bethe equations (22.25), by the Cartan matrix of the same diagram.

Spectral curves. § 10.5’s curve is a reduction too, of a much larger linear problem, and the modern statement — that integrable systems of this type are classified by moduli of curves with data on them — is the subject [Lecture 10](#)’s pointer was aiming at.

What all three share is the shape [Lecture 12](#) printed: a free system, a symmetry, a constraint and a choice of gauge. It is worth carrying out of the course as a question to ask of any solvable model — what is it a reduction of?

Summary.

- *Nothing relaxes; local observables do* ([Proposition 23.1](#)): the time average of a local observable is its value in the initial state dephased in energy. Eigenstate thermalisation ([23.2](#)) would make that a function of the energy alone; in an integrable model the *typical* eigenstate obeys it and the *worst* does not ([Figure 23.1](#)), because the other charges remember what was prepared.
- *The generalised Gibbs ensemble* ([23.3](#)) is the maximum-entropy state carrying every charge’s initial value, and for a transverse-field Ising quench the calculation is exact ([Theorem 23.6](#)): the occupations $n_k = \sin^2 \chi_k$ of the *difference* Bogoliubov angle are conserved, $m^z(t)$ is one Brillouin-zone integral ([23.6](#)), and its oscillating part decays like $t^{-3/2}$ — from a double zero of the amplitude at both stationary points of ε , which needs $h_0 \neq J$ (else t^{-1}) and $h > 0$ (else no decay at all), while $h_0 > 0$ picks the initial ground state out of a degenerate pair and $h \neq J$ is the proof’s route rather than the conclusion’s ([Remark 23.7](#)). The limit is the GGE value, not the Gibbs value of the same energy, by 0.134 here.
- *The limits do not commute* ([Remark 23.8](#)): at finite N the signal revives and only its time average converges. And an observable proportional to a conserved charge cannot tell the two ensembles apart at all ([Remark 23.7](#)).
- *Which combinations of the n_k are charges* ([Proposition 23.10](#)): away from $h = J$, $Q[f]$ is strictly local exactly when $f = \varepsilon \times$ a trigonometric polynomial. Without the ε the coefficients decay exponentially instead of stopping — a *quasi-local* charge — and for interacting models those are needed to close the ensemble.
- *One function per species carries the state*: the ensemble is a filling fraction $\vartheta(\lambda)$ and the charges are integrals ([23.11](#)) against it. Let ϑ depend on position and time and ([23.15](#)) follows — one continuity equation per rapidity, at [Lecture 19](#)’s dressed velocity ([23.14](#)), which must be a ratio of dressed derivatives and not $\varepsilon'/2\pi\rho_t$ outside thermal

equilibrium. For hard rods it is $(\lambda - a_J)/(1 - a_N)$ in one line (Proposition 23.13), and the rods themselves agree (Figure 23.4). The solution is the *pushforward* (23.16), with the inverse flow and the Jacobian.

- *The cradle*: a harmonic trap leaves the Euler-scale picture standing and, in the two free limits, costs nothing at all — the rapidity distribution turns and returns — while at intermediate coupling it drifts and the tube’s finite width breaks integrability outright. Weak breaking gives a prethermal plateau (Remark 23.15), the quantum relative of KAM, and whether it has a theorem behind it is open.
- *The dictionary, whole*: twenty-two rows, five theorems, all five Lecture 17’s and four of them the quantum Toda chain. The table is the subject.

Tutorial. Exercise Sheet 23.

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